0.1 Subspace Topology

Definition 0.1.1. Let $(X, \mathcal{T})$ be a Topological Space, and $A \subset X$ be a subset. Define a *Subspace Topology* on A as:

$$\mathcal{T}_A \overset{\text{def}}{=} \{U \cap A \mid U \in \mathcal{T}\}$$

Then, $\mathcal{T}_A$ is a Topology for A.

Lemma 0.1.2. Let $(X, \mathcal{T})$ be a Topological Space, and $A \subset X$ be a subset.

1. $V \subset A$ is open set of A *if and only if* For some open set U of X, $V = U \cap A$
2. $F \subset A$ is closed set of A *if and only if* For some closed set C of X, $F = C \cap A$

Proof. The first assertion is trivial. Let $F \subset A$ be a closed set of A. Then, $A \setminus F$ is open set of A. By definition, $A \setminus F = U \cap A$ for some open U of X. Now, $C = X \setminus U$ is closed of X with

$$C \cap A = (X \setminus U) \cap A = A \setminus (U \cap A) = A \setminus (A \setminus F) = F$$

Conversely, suppose that C is closed set of X. Then, $F = C \cap A$ satisfies

$$A \setminus F = A \setminus (C \cap A) = A \setminus C = A \cap (X \setminus C)$$

Thus, F is closed in A. □

Lemma 0.1.3. Let $(A, \mathcal{T}_A)$ be a subspace of Topological Space $(X, \mathcal{T})$. Then, the *inclusion map*

$$i : A \to X : a \mapsto a$$

is Continuous one-to-one map.

Proof. Injective is clear. Continuous: Let $U \subseteq X$ be an open set. Then, $i^{-1}[U] = U \cap A \in \mathcal{T}_A$. □

Using functional notation, we can write that $\mathcal{T}_A = \{i^{-1}[U] \mid U \in \mathcal{T}\}$.

Proposition 0.1.4. Let $(A, \mathcal{T}_A)$ be a subspace of Topological Space $(X, \mathcal{T})$. Then,

The inclusion $i : A \to X$ is Open map *if and only if* $A \subset X$ is open set of X

and

The inclusion $i : A \to X$ is Closed map *if and only if* $A \subset X$ is closed set of X

Proof. Trivial. □

Theorem 0.1.5. Let X be a Topological Space, and $B \subset A \subset X$.

1. If B is an open set of the subspace A and A is an open set of X, then B is an open set of X.
2. If B is a closed set of the subspace A and A is a closed set of X, then B is an open set of X.

Proof. Trivial. □